

## Research Paper

# Short-term training for explosive strength causes neural and mechanical adaptations

Neale A. Tillin<sup>1,2</sup>, Matthew T. G. Pain<sup>2</sup> and Jonathan P. Folland<sup>2</sup>

<sup>1</sup>School of Sport, Exercise and Health Sciences, Loughborough University, Leicestershire, UK

<sup>2</sup>Department of Life Sciences, University of Roehampton, London, UK

This study investigated the neural and peripheral adaptations to short-term training for explosive force production. Ten men trained the knee extensors with unilateral explosive isometric contractions (1 s 'fast and hard') for 4 weeks. Before and after training, force was recorded at 50-ms intervals from force onset ( $F_{50}$ ,  $F_{100}$  and  $F_{150}$ ) during both voluntary and involuntary (supramaximal evoked octet; eight pulses at 300 Hz) explosive isometric contractions. Neural drive during the explosive voluntary contractions was measured with the ratio of voluntary/octet force, and average EMG normalized to the peak-to-peak M-wave of the three superficial quadriceps. Maximal voluntary force (MVF) was also measured, and ultrasonic images of the vastus lateralis were recorded during ramped contractions to assess muscle–tendon unit stiffness between 50 and 90% MVF. There was an increase in voluntary  $F_{50}$  (+54%),  $F_{100}$  (+15%) and  $F_{150}$  (+14%) and in octet  $F_{50}$  (+7%) and  $F_{100}$  (+10%). Voluntary  $F_{100}$  and  $F_{150}$ , and octet  $F_{50}$  and  $F_{100}$  increased proportionally with MVF (+11%). However, the increase in voluntary  $F_{50}$  was +37% even after normalization to MVF, and coincided with a 42% increase in both voluntary/octet force and agonist-normalized EMG over the first 50 ms. Muscle–tendon unit stiffness between 50 and 90% MVF also increased. In conclusion, enhanced agonist neural drive and MVF accounted for improved explosive voluntary force production in the early and late phases of the contraction, respectively. The increases in explosive octet force and muscle–tendon unit stiffness provide novel evidence of peripheral adaptations within merely 4 weeks of training for explosive force production.

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**Corresponding author** N. A. Tillin: Department of Life Sciences, University of Roehampton, Whitelands College, Holybourne Avenue, London SW15 4JD, UK. Email: neale.tillin@roehampton.ac.uk

Explosive muscular force production, often measured as the rate of force development (RFD), is considered functionally important in situations where the time to develop force is limited, such as restabilization of the body following a loss of balance or during explosive sports activities (Aagaard *et al.* 2002; de Ruiter *et al.* 2004; Suetta *et al.* 2004; Tillin *et al.* 2010). Explosive force production is thought to be influenced by several neural, mechanical and architectural parameters, including agonist neural drive, maximal strength, fascicle length, fascicle angle and muscle–tendon unit (MTU) stiffness (de Ruiter *et al.* 2004; Bojsen-Møller *et al.* 2005; Andersen & Aagaard, 2006; Blazevich *et al.* 2009). However, the influence of different strength-training modalities on explosive force production and its various determinants is unclear

and has implications for improving health and sports performance.

Training to enhance maximal voluntary force (MVF) production via sustained (>2 s) high-load contractions ('training for maximal strength') has proved successful at increasing this component of strength, but there is equivocal evidence for its effects on explosive force production (Rich & Cafarelli, 2000; Kubo *et al.* 2001a, 2007; Aagaard *et al.* 2002; Suetta *et al.* 2004; Blazevich *et al.* 2009; Andersen *et al.* 2010; Tillin *et al.* 2011). After 4 weeks of training for maximal strength, we observed no change in explosive force production or agonist neural drive over the first 150 ms of explosive contractions, despite an increase in both MVF and agonist neural drive at MVF (Tillin *et al.* 2011). This suggests that short-term improvements

in explosive and maximal force production may require distinct training stimuli that elicit different adaptations.

Training with short ( $\sim 1$  s) high-RFD contractions, which place a substantial emphasis on producing explosive rather than maximal force ('training for explosive strength'), may provide the specific stimuli required to improve this aspect of force production. Furthermore, the smaller loading volume associated with training for explosive strength and the possibility of gains in explosive force production after merely 4 weeks (Barry *et al.* 2005; Gruber *et al.* 2007; Geertsen *et al.* 2008) suggest that functional improvements may be obtained without the level of fatigue and discomfort typically associated with training for maximal strength. However, the neural, mechanical and architectural adaptations to training for explosive strength, and their functional effects, have not been widely investigated. Early evidence has linked RFD gains after training for explosive strength to increased motor unit firing rate (measured using fine-wire EMG; Van Cutsem *et al.* 1998) and agonist surface EMG in the early phase of explosive contractions (Barry *et al.* 2005; Gruber *et al.* 2007). However, the latter two surface EMG studies did not normalize recordings to a maximal M-wave, so their results may have been confounded by variations in signal conduction and electrode relocation (Gandevia, 2001; Folland & Williams, 2007). In contrast, a more recent study reported no change in agonist EMG normalized to a maximal M-wave, despite increases in RFD following explosive strength training (Geertsen *et al.* 2008). The variability inherent in surface EMG recordings may partly explain these equivocal results. An alternative method of measuring neural activation during explosive contractions might be provided by calculating voluntary force as a proportion of supramaximal evoked octet force (i.e. voluntary force in relation to the maximal capacity of the MTU for explosive force production evoked by eight pulses at 300 Hz; de Ruiter *et al.* 1999, 2004). However, this technique has not been employed in the context of training for explosive strength.

Whilst there is early evidence for increased neural drive following short-term training for explosive strength, it is unclear whether mechanical and architectural adaptations might also occur. Any changes in muscle size, architecture or MTU stiffness could potentially also contribute to improvements in voluntary and involuntary explosive force production after training. Previous studies of training for maximal strength have reported changes in muscle architecture after 4–5 weeks (Blazevich *et al.* 2007; Seynnes *et al.* 2007) and increased tendon stiffness after 9–10 weeks (Seynnes *et al.* 2009; Kubo *et al.* 2010), but the volume of loading in this situation is likely to be greater than that experienced when training purely for explosive force production. This might suggest that mechanical and architectural adaptations are unlikely to occur with short-term training purely for

explosive strength; however, this hypothesis has not yet been tested. Clearly, further investigation is required to understand the neuromuscular adaptations to training for explosive strength. The purpose of this study was to investigate changes in explosive force production after 4 weeks of training for explosive strength, rather than maximal force production, and provide a detailed assessment of the neural, mechanical and architectural adaptations contributing to any observed changes in function.

## Methods

### Participants

Ten male participants (age,  $20 \pm 2$  years; height,  $182 \pm 7$  cm; mass,  $74 \pm 7$  kg) completed the study. Participants were recreationally active (moderate exercise four times per week or less), but not involved in any form of strength or power training. All the participants were healthy, injury free and provided written informed consent prior to their involvement in this study, which was approved by the Loughborough University ethical advisory committee and complied with the provisions of the Declaration of Helsinki.

### Overview

Participants completed a familiarization session and two measurement sessions before (each separated by 2–3 days), and two measurement sessions after (separated by 2–3 days), 4 weeks of isometric explosive strength training of the knee extensors. Training was completed on one leg (trained leg) chosen at random, but all pre- and post-training measurements were taken from both the trained and the untrained legs. Pretraining measurement sessions took place within 10 days prior to the start of training, whilst post-training measurement sessions took place 2–3 (first measurement session) and 4–6 days after the last training session (second measurement session). During the familiarization session, participants completed the range of measurement tasks without data recording. All measurement sessions involved recording knee extension/flexion force and surface EMG (superficial knee extensors and bicep femoris) during isometric contractions. The first of the pre- and post-training measurement sessions involved electrically evoked twitch and octet contractions, explosive voluntary contractions and maximal voluntary contractions (MVCs) of the knee extensors. The second of the pre- and post-training measurement sessions involved recording ultrasonic images of the vastus lateralis at rest and during ramped MVCs of the knee extensors in order to assess mechanical and architectural properties of the MTU.

## Training

The aim of the training programme was to enhance explosive rather than maximal force production, whilst minimizing the volume of loading and fatigue. Participants completed four training sessions per week for 4 weeks. Each session consisted of a brief warm up of submaximal isometric knee extensions, followed by four sets (separated by 2 min) of 10 unilateral explosive isometric knee extensions (separated by 5 s) using the same apparatus as the measurement trials. Prior to each contraction, participants were instructed to relax completely before pushing 'fast and hard' for ~1 s, in an attempt to achieve at least 90% of their MVF as quickly as possible, and without prior countermovement. Maximal voluntary force was first determined (via the method detailed in the 'Maximal voluntary contractions' subsection) in the familiarization session and then redetermined in the first pretraining measurement session and at the start of the first training session each week. Participants were instructed to avoid contracting the untrained leg during each effort. For biofeedback, a computer monitor displayed the force–time curve with a cursor at 90% of MVF, the slope of the force–time curve (established with a 1-ms constant epoch) and the baseline of the force–time curve. The baseline was used to confirm that a countermovement had not occurred.

## Force and EMG recording

All measurement and training sessions were completed in an isometric strength-testing chair (Parker *et al.* 1990; Bojsen-Møller *et al.* 2005), with knee and hip angles of 85 and 100 deg, respectively (180 deg representing full extension). A waist belt and shoulder straps firmly secured participants in the chair. Force was measured via a linear strain gauge positioned perpendicular to the tibia (Tillin *et al.* 2010, 2011). To measure EMG, single differential surface electrodes (Delsys Bagnoli-4; Delsys, Boston, MA, USA) were placed at the following anatomical locations: over the belly of the rectus femoris (RF), vastus lateralis (VL), vastus medialis (VM) and biceps femoris (BF); parallel to the presumed orientation of the muscle fibres; and at ~50 (RF), 55 (VL), 90 (VM) and 45% (BF) of the distance between the greater trochanter and the lateral femoral condyle. The location of each electrode during pretesting was outlined on the skin surface with a permanent marker, and marks were reapplied on each successive visit to the laboratory (i.e. every 2–3 days) to ensure identical placement post-testing. Force and EMG signals were amplified (force,  $\times 500$ ; EMG,  $\times 100$ ; differential amplifier 20–450 Hz), and sampled at 2000 Hz with an analog-to-digital convertor (CED micro 1401; CED, Cambridge, UK) and PC utilizing Spike 2 software (CED). Force was digitally notch filtered at 100 and

200 Hz (q-factor of 100; Tillin *et al.* 2010, 2011), whilst EMG signals were bandpass filtered (6–500 Hz) in both directions using a second-order Butterworth digital filter.

For measurements of MTU stiffness (see section 'Second measurement session'), internal quadriceps force was estimated using previously reported methods (Bojsen-Møller *et al.* 2005). Briefly, measured force was corrected for the negative contribution from antagonist coactivation as follows. Root mean square (RMS) of the antagonist (BF) EMG signal (determined via a 250-ms moving epoch) at a given sampled time point during the ramped contraction, as a ratio of maximal BF RMS EMG (see section 'Maximal voluntary contractions'), was multiplied by maximal knee flexor force and summed with measured extensor force at the same sampled time point. Corrected extensor force was then multiplied by the external moment arm (distance along the tibia between the anterior edge of the tibial condyle and the centre of the ankle strap) to provide an external moment of force. Finally, the external moment of force was divided by the internal knee moment arm, derived as a function of hip and knee angle from Visser *et al.* (1990), to provide internal quadriceps force.

## First measurement session (pre- and post-training)

Measurements detailed below were completed first with one leg and then repeated with the other leg. A consistent, but randomly assigned, order was used for each individual.

**Electrically evoked contractions.** The femoral nerve was electrically stimulated (DS7AH; Digitimer Ltd, Hertfordshire, UK) with square-wave pulses (0.1 ms duration) to elicit compound muscle action potentials (M-waves), twitch contractions and octet contractions. The anode (carbon rubber, 7 cm  $\times$  10 cm; EMS Physio Ltd, Greenham, UK) was taped to the skin over the greater trochanter. The cathode, a custom-adapted stimulation probe (1 cm diameter; Electro Medical Supplies, Wantage, UK) protruding 2 cm perpendicular from the centre of a plastic base (4 cm  $\times$  5 cm), was taped to the skin over the femoral nerve in the femoral triangle. The precise location of the cathode was determined as the position that evoked the greatest twitch response for a particular submaximal electrical current (typically 30–50 mA).

After establishing the maximal knee extensor twitch response for a single electrical impulse (Tillin *et al.* 2010, 2011), three supramaximal (120%) twitches were elicited at 12-s intervals. Variables recorded and averaged across the three supramaximal twitch contractions included the following: peak-to-peak M-wave ( $M_{\max}$ ) response of the RF, VL and VM; peak force; time to peak force; force at 50 ms from force onset ( $F_{50}$ ); and involuntary maximal electromechanical delay ( $EMD_{\max}$ ; defined as the time difference between the first agonist EMG onset

and force onset). Three octet contractions (eight pulses at 300 Hz) were also evoked at the same supramaximal current as the twitch contractions. Variables recorded and averaged across the three supramaximal octet contractions included the following: peak force; and force at 50 and 100 ms from force onset ( $F_{50}$  and  $F_{100}$ ). Octet forces were also normalized to MVF. Octet data from only seven participants were obtained as a result of the discomfort caused by this type of stimulation.

**Explosive voluntary contractions.** Following the evoked contractions, participants completed a warm up of submaximal voluntary contractions, followed by 10 explosive voluntary contractions (each  $\sim 20$  s apart). Prior to each contraction, participants were provided with the same instruction as that given during the training efforts. Explosive voluntary contractions were completed separately from MVCs (detailed in the ‘*Maximal voluntary contractions*’ subsection) because previous work has highlighted the importance of instruction with regard to performance outcome (Sahaly *et al.* 2001). The three explosive voluntary contractions with the largest peak slope and no change in baseline force  $>0.5$  N during the 100 ms prior to contraction onset were used for analysis. Voluntary explosive force was measured at 50, 100 and 150 ms from force onset ( $F_{50}$ ,  $F_{100}$  and  $F_{150}$ ), and these values are reported in absolute terms and relative to MVF. Voluntary  $F_{50}$  and  $F_{100}$  were also calculated as a percentage of octet  $F_{50}$  and  $F_{100}$ , respectively (voluntary/octet), to provide a measure of agonist neural drive. Furthermore, the RMS EMG of each muscle was measured in three 50 ms time windows ( $EMG_{0-50}$ ,  $EMG_{50-100}$  and  $EMG_{100-150}$ ) from EMG onset (first agonist muscle to be activated). Agonist (RF, VL and VM) RMS EMG values were normalized to  $M_{\max}$  and averaged across the three muscles to give a mean agonist value. Antagonist (BF) RMS EMG values were normalized to maximal BF RMS EMG determined during the knee flexor MVCs (detailed in the ‘*Maximal voluntary contractions*’ subsection). Voluntary  $EMD_{\max}$  was defined as the time difference between EMG and force onset and is reported in absolute terms and as a percentage of involuntary  $EMD_{\max}$ . All explosive voluntary force and EMG variables were averaged across the three explosive voluntary contractions chosen for analysis.

Force and EMG onsets during the explosive voluntary and evoked contractions were identified manually, which is considered the ‘gold standard’ method for detecting signal onsets (Allison, 2003; Moretti *et al.* 2003; Pain & Hibbs, 2007; Pulkovski *et al.* 2008). Details of this method and its reliability have previously been published (Tillin *et al.* 2010). Briefly, the same investigator viewed signal recordings with a constant  $y$ -axis scale of 1 N and 10 mV, for force and EMG respectively, with an  $x$ -axis scale initially of 500 ms and then of 25 ms for determining signal onset. This provided a good resolution from which

to establish the pattern of the noise and interpolate signal onset, defined as the last peak/trough before the signal deflected away from the baseline.

**Maximal voluntary contractions.** Following the explosive contractions, participants completed three knee extensor isometric MVCs (separated by  $\geq 30$  s), in which participants were instructed to push as hard as possible for 3–5 s. Biofeedback and verbal encouragement were provided during and between each MVC. Knee extensor MVF was the greatest voluntary force achieved by a participant in any of the knee extensor MVCs or explosive contractions during that laboratory visit. The RMS EMG of each agonist muscle was recorded during a 500 ms epoch at MVF (250 ms either side of MVF), normalized to  $M_{\max}$  and averaged across the three quadriceps muscles. Participants then completed three knee flexor isometric MVCs at the same hip and knee joint angles as the knee extensor contractions. The RMS EMG during a 500 ms epoch at the greatest knee flexor force (250 ms either side) was measured to give maximal BF RMS EMG.

## Second measurement session (pre- and post-training)

An ultrasound machine (Toshiba, Power Vision 6000, SSA-370A; Mount International Ultrasound Services Ltd) with a linear array probe (7.5 MHz B-mode, scanning width 60 mm and depth 50 mm) was used to collect video images of the VL. The probe was held longitudinally over the VL at 50% of the distance between the greater trochanter and the lateral femoral condyle. An echo-absorbent marker (elastic band), taped over the skin in the axial plane, provided a reference point within the video images of any probe movement relative to the skin. The probe position during the pretraining measurement session was outlined on the skin with permanent marker, and these marks were reapplied during each visit to the laboratory (i.e. every 2–3 days) to ensure identical placement post-testing. Images were recorded with a digital video recorder (Sony Walkman, GV-D900E) at 25 Hz and synchronized with force recordings via a custom-built remote trigger. The trigger generated a pulse on the force trace and a simultaneous high-frequency sound wave, which was recorded on the digital video with the ultrasonic images. During offline analysis, the processed force and EMG traces were exported at 25 Hz to match the sampling frequency of the video. Video recordings were imported into manual digitization software (with a resolution of 0.16 mm; Avi Digitiser Version 1.2 Beta; RF Spectrum Modelling Ltd, Swindon, UK) for further analysis. This digitization software enabled identification of the first frame during which the high-frequency sound wave occurred.

Ultrasound images were recorded while participants were at rest and then throughout two ramped knee extensor MVCs (separated by  $\sim 2$  min), in which

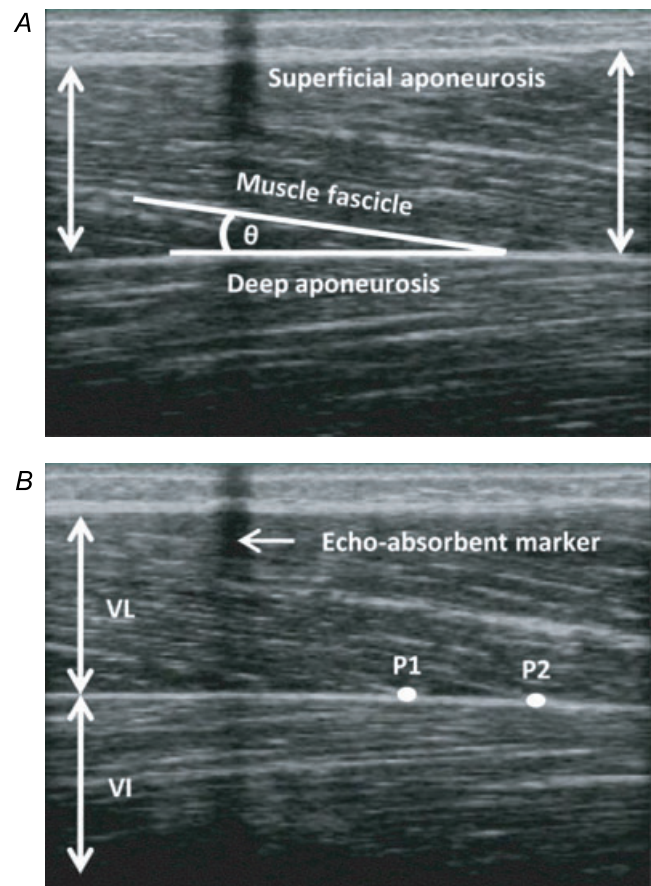
participants steadily increased force from rest to MVF over an 8-s period. Participants were asked to perform another effort if peak force differed between the two ramped contractions by more than 10%. Participants then completed three knee flexor MVCs to establish the greatest flexor force and maximal BF RMS EMG as outlined above. The video image providing the clearest definition of muscle fascicles and deep aponeurosis of the two ramped knee extensor MVCs was used for further analysis. Muscle thickness, fascicle angle and fascicle length were measured in 10 consecutive frames recorded at rest, and averaged across these 10 frames (Fig. 1A). Muscle thickness was defined as the mean distance between the superficial and deep aponeurosis measured at both ends of the 60 mm image. Fascicle angle was defined as the angle between a muscle fascicle, chosen at random, and the deep aponeurosis. The full length of the muscle fascicles was not visible in the images, so fascicle length was extrapolated, using trigonometry, from fascicle angle and muscle thickness. Errors associated with this extrapolation method are typically within the region of 2–7% (Finni & Komi, 2002; Finni *et al.* 2003; Ishikawa *et al.* 2003).

To assess MTU stiffness, the displacement of two visible cross-points between a muscle fascicle and the aponeurosis, relative to the echo-absorbent marker, were tracked and digitized throughout the ramped MVC with the clearest images (Fig. 1B). The average displacement of the two cross-points across five consecutive frames was recorded at rest and at 10% increments of internal quadriceps force up to internal quadriceps MVF; i.e. one frame at, and two frames either side of, each 10% increment. The relationship between aponeurosis displacement and internal quadriceps force was plotted, and the slope of this relationship from 10 to 50% and 50 to 90% of internal quadriceps MVF was defined as MTU stiffness at low and high force levels, respectively.

Soft tissue compression during isometric knee extensor MVCs causes knee angle changes and subsequent passive aponeurosis displacement (Bojsen-Møller *et al.* 2003; Stafiliadis *et al.* 2005; Arampatzis *et al.* 2007). We were unable to correct for passive aponeurosis displacement owing to erroneous electro-goniometer data collected during the measurement sessions. However, during extensive pilot work we found changes in knee angle to be <4 deg during ramped MVCs, suggesting that there was minimal system compliance in our model. Furthermore, we believe that for this repeated-measures study design any changes in soft tissue compression pre- to post-training would be minimal, and the within-participant comparison of trained *versus* untrained leg would be expected to control for any possible changes in passive aponeurosis displacement. Mechanical variables measured for one participant using ultrasound were not included in the statistical analysis, because his deep aponeurosis was poorly defined in the video images.

## Data analysis and statistics

The reliability of mechanical and architectural variables (including twitch and octet force characteristics, MTU stiffness, muscle thickness, fascicle angle and fascicle length) was assessed by calculating intraclass correlation coefficients (ICCs) for repeated measures (i.e. pretraining and post-training) in the untrained leg. To investigate the coefficient of variation in mechanical variables as a result of manual digitization errors, six ultrasound recordings were reanalysed by the same investigator a week after the original analysis. Earlier investigations by our group have



**Figure 1.** Ultrasonic images of the vastus lateralis (VL) and vastus intermedius (VI) muscles at rest (A) and during a maximal voluntary isometric knee extension (B)

A, VL muscle thickness (mean distance between superficial and deep aponeurosis at both ends of the image), fascicle angle ( $\theta$ ; angle between a muscle fascicle and the deep aponeurosis) and fascicle length (extrapolated using trigonometry) were determined at rest. B, the displacement of two cross-points between VL muscle fascicles and the deep aponeurosis (P1 and P2), relative to the echo-absorbent marker, was tracked throughout ramped isometric knee extensions up to the maximal voluntary effort. The average displacement of these cross-points was plotted against internal quadriceps force to measure muscle–tendon stiffness.

reported good repeatability of the volitional parameters assessed in this study (Tillin *et al.* 2010, 2011).

Absolute values are reported as means  $\pm$  SD, and data for each dependent variable were analysed with a two-way repeated-measures ANOVA [time (pre *versus* post)  $\times$  leg (trained *versus* untrained)]. In the incidence of a main effect ( $P < 0.05$ ) or a tendency for a main effect ( $P \leq 0.06$ ), Student's paired *t* tests were used to determine within-leg differences between pre- and post-training measures, and a stepwise Bonferroni correction procedure was employed for all time series data. Statistical analyses were completed using SPSS version 16 (SPSS Inc., Chicago, IL, USA), and the significance level was set at  $P < 0.05$ . Percentage changes of dependent variables are presented in order to show the magnitude of significant effects, and were calculated as the percentage change of the mean. Pearson's product-moment correlations were used to assess the strength of bivariate relationships between changes in key dependent variables.

## Results

### Voluntary force

Maximal voluntary force increased by 11% in the trained leg (pre,  $482 \pm 58$  *versus* post,  $535 \pm 61$  N;  $P < 0.001$ ) and by 4% in the untrained leg (pre,  $489 \pm 58$  *versus* post,  $508 \pm 62$  N;  $P = 0.007$ ). This increase was greater in the trained leg, resulting in a time-by-leg interaction effect ( $P = 0.002$ ).

During the explosive voluntary contractions there was also a time-by-leg interaction effect on absolute  $F_{50}$ ,  $F_{100}$  and  $F_{150}$  ( $0.004 \leq P \leq 0.013$ ). Paired comparisons revealed that  $F_{50}$  (+54%;  $P = 0.009$ ),  $F_{100}$  (+15%;  $P = 0.004$ ) and  $F_{150}$  (+14%;  $P < 0.001$ ) increased in the trained leg, but were unchanged in the untrained leg (Fig. 2A and B). Normalized  $F_{50}$  (normalized to MVF) increased in the trained leg (+37%;  $P = 0.044$ ), but not in the untrained leg (Fig. 2C and D), reflecting a time-by-leg interaction effect ( $P = 0.016$ ). Training did not affect normalized

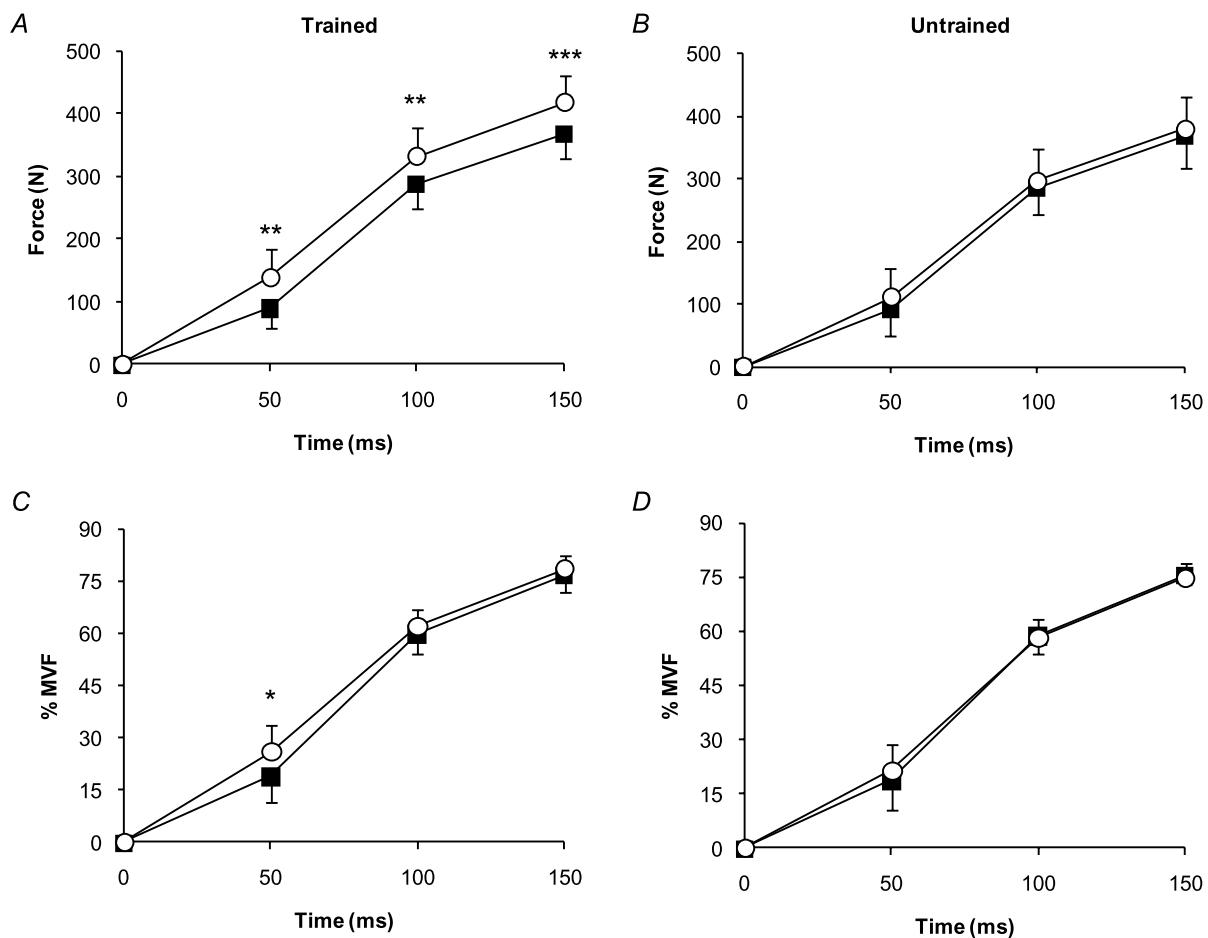


Figure 2. Absolute (A and B) and normalized force [C and D; normalized to maximal voluntary force (MVF)] during explosive isometric knee extensions, pre- (filled squares) and post-training (open circles) for explosive strength in the trained (A and C) and untrained legs (B and D). Data are means  $\pm$  SD ( $n = 10$ ). A training effect is denoted by \* ( $P < 0.05$ ), \*\* ( $P < 0.01$ ) or \*\*\* ( $P < 0.001$ ).

**Table 1. Evoked force parameters recorded in the trained and untrained legs during supramaximal octet and twitch contractions of the knee extensors, pre- and post-training for explosive strength**

| Evoked force parameter |                         | Trained leg |            | Untrained leg |          | ICC    |
|------------------------|-------------------------|-------------|------------|---------------|----------|--------|
|                        |                         | Pre         | Post       | Pre           | Post     |        |
| Octet                  | Peak force (N)          | 320 ± 31    | 353 ± 18** | 320 ± 48      | 341 ± 36 | 0.85*  |
|                        | Force at 50 ms (N)      | 193 ± 11    | 207 ± 13*  | 199 ± 30      | 205 ± 21 | 0.92** |
|                        | Force at 100 ms (N)     | 301 ± 31    | 330 ± 19** | 303 ± 50      | 320 ± 37 | 0.86*  |
|                        | Force at 50 ms (% MVF)  | 40 ± 4      | 39 ± 4     | 40 ± 2        | 41 ± 3   | —      |
|                        | Force at 100 ms (% MVF) | 62 ± 6      | 62 ± 7     | 61 ± 4        | 64 ± 6   | —      |
| Twitch                 | Peak force (N)          | 128 ± 25    | 132 ± 15   | 126 ± 20      | 133 ± 14 | 0.82** |
|                        | Time to peak force (ms) | 87 ± 8      | 86 ± 6     | 84 ± 5        | 85 ± 5   | 0.91** |
|                        | Force at 50 ms (N)      | 99 ± 14     | 102 ± 11   | 101 ± 14      | 103 ± 11 | 0.83** |

Abbreviation: MVF, maximal voluntary force. The intra-class correlation coefficients (ICC) are shown for the repeated absolute measurements of the untrained leg. Data are mean ± SD (octet,  $n = 7$ ; twitch,  $n = 10$ ). A training effect or significant ICC is denoted by \* ( $P < 0.05$ ) or \*\* ( $P < 0.01$ ).

$F_{100}$  or  $F_{150}$  in either the trained or the untrained leg (Fig. 2C and D).

### Evoked force

There was a main effect of time on peak force,  $F_{50}$ , and  $F_{100}$  during the evoked octet contractions ( $0.01 \leq P \leq 0.044$ ). Paired comparisons of the trained leg revealed an increase in octet peak force (+10%;  $P = 0.009$ ),  $F_{50}$  (+7%;  $P = 0.039$ ), and  $F_{100}$  (+10%;  $P = 0.018$ ) following training (Table 1). However, octet  $F_{50}$  and  $F_{100}$  normalized to MVF were unchanged in the trained leg following training (Table 1). This reflected the significant correlations between the changes in MVF and the changes in both octet  $F_{50}$  ( $r = 0.80$ ;  $P = 0.031$ ) and  $F_{100}$  ( $r = 0.83$ ;  $P = 0.022$ ). The untrained leg showed unchanged mechanical responses to the octet stimulation following training (Table 1). Twitch peak force, time to peak force and  $F_{50}$  were unaffected by training in both the trained and the untrained legs (Table 1).

### Neural drive

There were no main effects of the training on agonist-normalized EMG at MVF or  $\text{EMG}_{100-150}$  during the explosive voluntary contractions. In contrast, there was a main effect of time on agonist-normalized  $\text{EMG}_{50-100}$  ( $P = 0.048$ ) and a tendency for a main effect of time on  $\text{EMG}_{0-50}$  ( $P = 0.06$ ). Paired comparisons in the trained leg showed a 42% increase in agonist-normalized  $\text{EMG}_{0-50}$  ( $P = 0.022$ ), but no change in  $\text{EMG}_{50-100}$  (Fig. 3A). Training did not affect agonist-normalized EMG in the untrained leg during the explosive voluntary contractions (Fig. 3B).

There was an increase in voluntary/octet  $F_{50}$  in the trained (+42%  $P = 0.037$ ; Fig. 4A), but not in the untrained leg (Fig. 4B), reflecting a time-by-leg interaction effect ( $P = 0.016$ ). There was also a time-by-leg interaction

effect on voluntary/octet  $F_{100}$  ( $P = 0.047$ ), but paired comparisons showed no change in this variable in either the trained or the untrained leg following training (Fig. 4). The individual changes in voluntary/octet  $F_{50}$  and  $F_{100}$  in the trained leg were very highly correlated with the changes in normalized voluntary  $F_{50}$  ( $r = 0.99$ ;  $P < 0.001$ ) and  $F_{100}$  ( $r = 0.87$ ;  $P = 0.011$ ), respectively.

Antagonist-normalized EMG during the explosive contractions was also similar pre- and post-training for both the trained and the untrained legs. Training did not affect voluntary  $\text{EMD}_{\text{max}}$  (~16 ms in both legs and conditions), involuntary  $\text{EMD}_{\text{max}}$  (~7 ms in both legs and conditions) or voluntary  $\text{EMD}_{\text{max}}$  relative to involuntary  $\text{EMD}_{\text{max}}$  (~230% in both legs and conditions).

### Mechanical and architectural properties of the MTU

There was a time-by-leg interaction effect on MTU stiffness at high force levels (50–90% internal quadriceps MVF;  $P = 0.003$ ), resulting from an increase in the trained leg (+34%;  $P = 0.002$ ), but not the untrained leg, following training (Table 2). There were, however, no main effects on MTU stiffness at low force levels (10–50% internal quadriceps MVF; Table 2).

There was a main effect of time on muscle thickness ( $P = 0.024$ ), and paired comparisons revealed a tendency for an increase in muscle thickness in the trained leg (+7%;  $P = 0.054$ ), but not in the untrained leg (Table 2). There were no effects of training on fascicle length or fascicle angle (Table 2). The coefficients of variation for the repeated analysis of the same ultrasound recordings were as follows: MTU stiffness between 50 and 90% internal quadriceps MVF, 8.3%; muscle thickness, 0.2%; fascicle angle, 3.6%; and fascicle length, 3.6%.

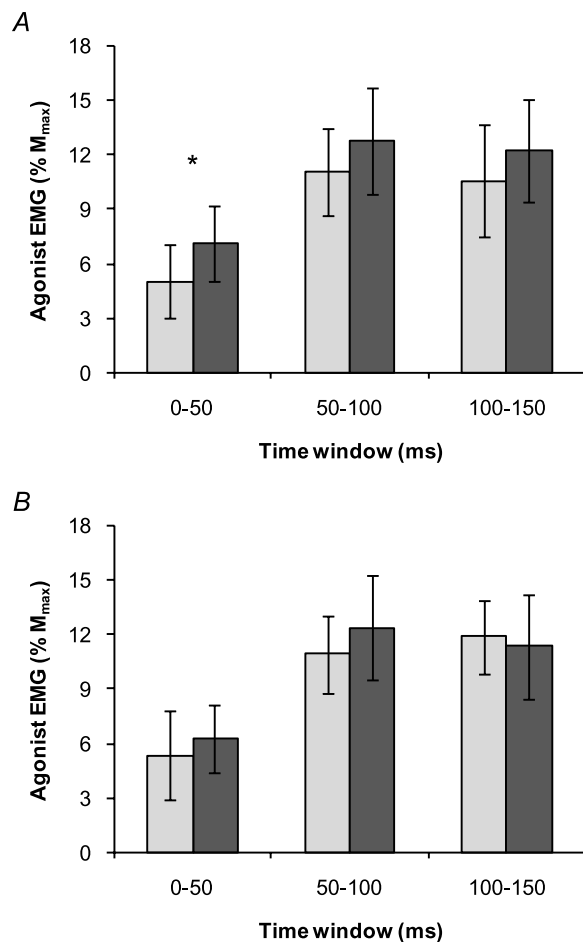
### Discussion

The purpose of this study was to investigate the neuromuscular responses to 4 weeks of training for

explosive, rather than maximal, force production. Results showed an increase in explosive voluntary force production at all measured time points from force onset. The increase in voluntary  $F_{50}$  appeared to be primarily due to enhanced agonist neural drive in the same time period, whilst the increase in voluntary  $F_{100}$  and  $F_{150}$  were probably due to the proportional increase in MVF. The increase in octet explosive force demonstrated an enhanced capacity of the MTU for explosive force production and indicates that peripheral adaptations also occurred as a result of the training. Peripheral adaptations included increased MTU stiffness measured between 50 and 90% internal quadriceps MVF; however, this is unlikely to explain the increased octet explosive force,

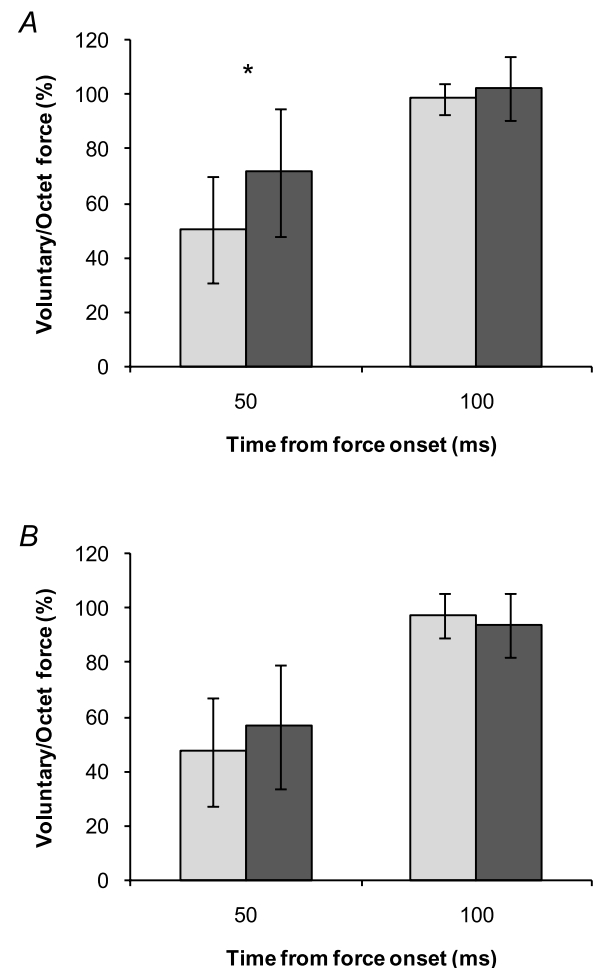
which was measured over a different range of forces (0–62% MVF). Instead, changes in octet explosive force were proportional to, and correlated with, changes in MVF, suggesting a common peripheral mechanism for these changes.

Voluntary explosive force at 50, 100 and 150 ms after force onset increased in the trained leg, primarily due to the greater RFD during the initial 50 ms of contraction. Three other studies have observed increased RFD after merely 4 weeks of explosive strength training (Barry *et al.* 2005; Gruber *et al.* 2007; Geertsens *et al.* 2008). In contrast, we previously found that 4 weeks of training for maximal strength, without a specific emphasis on performing the contractions explosively, had no effect on RFD (Tillin *et al.* 2011). Moreover, there is equivocal evidence for increased



**Figure 3.** Agonist-normalized EMG of the trained (A) and untrained legs (B) during explosive isometric knee extensions performed pre- (light bars) and post-training for explosive strength (dark bars)

Agonist-normalized EMG was measured with the root mean square of the EMG signal normalized to maximal M-wave (%  $M_{max}$ ) and analysed in three 50 ms time windows (0–50, 50–100 and 100–150 ms). Values were averaged across the three superficial quadriceps muscles, and group data are expressed as means  $\pm$  SD ( $n = 10$ ). A training effect is denoted by \* ( $P < 0.05$ ).



**Figure 4.** Force at 50 and 100 ms from force onset during explosive voluntary isometric knee extensions, as a percentage of octet force at 50 and 100 ms, respectively (voluntary/octet), pre- (light bars) and post-training for explosive strength (dark bars) in the trained (A) and untrained legs (B)

Data are means  $\pm$  SD ( $n = 7$ ). A training effect is denoted by \* ( $P < 0.05$ ).

**Table 2. Mechanical and architectural properties of the muscle–tendon unit (MTU) of the trained and untrained legs, measured at the vastus lateralis pre- and post-training for explosive strength**

| Parameter                                          | Trained leg |               | Untrained leg |               | ICC    |
|----------------------------------------------------|-------------|---------------|---------------|---------------|--------|
|                                                    | Pretraining | Post-training | Pretraining   | Post-training |        |
| MTU stiffness at 10–50% MVF ( $\text{N mm}^{-1}$ ) | 460 ± 149   | 399 ± 130     | 456 ± 175     | 423 ± 131     | 0.29   |
| MTU stiffness at 50–90% MVF ( $\text{N mm}^{-1}$ ) | 520 ± 86    | 697 ± 103**   | 592 ± 131     | 617 ± 186     | 0.88** |
| Muscle thickness (mm)                              | 25.8 ± 3.5  | 27.5 ± 4.1    | 26.0 ± 2.7    | 27.0 ± 3.34   | 0.93** |
| Fascicle angle (deg)                               | 17.4 ± 3.2  | 18.0 ± 3.9    | 16.8 ± 2.8    | 17.3 ± 2.7    | 0.92** |
| Fascicle length (mm)                               | 87.6 ± 9.6  | 91.8 ± 12.2   | 92.5 ± 16.2   | 92.9 ± 14.3   | 0.92** |

Abbreviation: MVF, internal quadriceps maximal voluntary force. The intraclass correlation coefficients (ICCs) are shown for the repeated measurements of the untrained leg. Data are means ± SD ( $n = 9$ ). A training effect or significant ICC is denoted by \* ( $P < 0.05$ ) or \*\* ( $P < 0.01$ ).

RFD with training for maximal strength over periods longer than 4 weeks (Kubo *et al.* 2001a, 2007; Aagaard *et al.* 2002; Suetta *et al.* 2004; Pucci *et al.* 2006; Blazevich *et al.* 2009; Andersen *et al.* 2010). **It therefore appears that short-duration contractions performed with the emphasis on producing explosive, rather than maximal, force may provide a more effective stimulus for improving explosive force/RFD than sustained high-load contractions. This has potential implications for training prescription, as the smaller loading volume experienced with purely explosive strength training may evoke substantial functional benefits (increased explosive and maximal strength), without the discomfort and fatigue associated with maximal strength training.**

When normalized to MVF,  $F_{50}$  increased following training (+37%) but  $F_{100}$  and  $F_{150}$  were unchanged. Consequently, the 11% gains in maximal strength accounted for improved force production during the late phase ( $\geq 100$  ms), but not during the early phase (first 50 ms) of explosive contractions. This is consistent with the increased influence of MVF in the later phase of explosive force production (Andersen & Aagaard, 2006). The differential response of early and late-phase normalized explosive voluntary force in the trained leg is remarkably similar to the difference we recently observed between explosive power athletes and untrained individuals (i.e. normalized  $F_{50}$  was greater in the athletes, but normalized  $F_{100}$  and  $F_{150}$  were similar between the two groups; Tillin *et al.* 2010).

The increase in early normalized explosive voluntary force appears to have resulted from neural adaptations specific to explosive force production. The 42% increase in voluntary/octet  $F_{50}$  indicates an improved ability to utilize the capacity of the MTU for explosive force production, and the similar 42% increase in normalized agonist  $\text{EMG}_{0-50}$  (normalized to  $M_{\text{max}}$ ) supports a change in neural drive during this early period. Earlier studies have reported equivocal evidence for increased agonist activation (determined via surface EMG measurements) after 4 weeks of training for explosive strength (Barry *et al.* 2005; Gruber *et al.* 2007; Geertsens *et al.* 2008),

but two of these studies (Barry *et al.* 2005; Gruber *et al.* 2007) did not normalize EMG to  $M_{\text{max}}$ , and none of them measured voluntary/octet force. The results of the present study therefore provide unequivocal evidence of enhanced agonist neural drive in the early phase of explosive voluntary contractions, following short-term training for explosive strength. Specific neural adaptations may include increased firing frequency and/or incidence of doublet discharge (which have been reported after 12 weeks of explosive strength training; Van Cutsem *et al.* 1998) and/or increased motor unit synchrony in the early phase of explosive contractions.

Measures of neural drive during the later phase of explosive contractions (i.e. normalized agonist  $\text{EMG}_{50-100}$ ,  $\text{EMG}_{100-150}$  and voluntary/octet  $F_{100}$ ) and at MVF (i.e. normalized agonist EMG) were unchanged after training. This suggests that peripheral adaptations were the primary mechanism for the improvements in MVF and later phase explosive voluntary force. Peripheral adaptations were assessed via supramaximal octet contractions, and results showed an increase in octet  $F_{50}$  and  $F_{100}$ . This is a unique finding and reflects an increased capacity of the MTU for explosive force production after explosive strength training. Interestingly, octet  $F_{50}$  and  $F_{100}$  increased in proportion to MVF (i.e. normalized octet force was unchanged), and these changes were highly correlated ( $r \leq 0.80$ ), suggesting a common peripheral explanation.

Despite the short duration of this training study and the limited loading volume of the explosive contractions, the increased muscle thickness (which approached significance;  $P = 0.054$ ; +7%) provides evidence of a hypertrophic response to the training. Hypertrophy could therefore be a common peripheral mechanism for the increased capacity of the MTU for both explosive and maximal force production. Whilst hypertrophy has been observed after only 3 weeks with maximal strength training (Seynnes *et al.* 2007), this is the first study to suggest a similar effect with explosive strength training. Further investigation, possibly with a larger sample size and using more sensitive methods to measure hypertrophic adaptations (i.e. magnetic

resonance imaging) is required to corroborate these findings.

The increase in MTU stiffness at high force levels after only 16 training sessions occurred 4–5 weeks earlier than typically expected with maximal strength training (Kubo *et al.* 2001a, 2006, 2007, 2010; Reeves *et al.* 2003; Seynnes *et al.* 2009). This is a surprising observation, considering that mechanical strain magnitude and volume are likely to be lower with explosive, as opposed to maximal, strength training, and both parameters have a positive influence on the biochemical (Arnoczky *et al.* 2002) and biomechanical response of connective tissue to loading (Kubo *et al.* 2001b; Yamamoto *et al.* 2005). However, explosive contractions provide a greater mechanical strain rate which, due to the viscoelastic properties of the MTU, predisposes the system to greater mechanical stress and strain energy for the same mechanical strain (McElhaney, 1966), and could potentially explain the rapid changes in MTU stiffness we have observed. Although MTU stiffness is thought to have a positive influence on RFD (Wilson *et al.* 1994; Bojsen-Møller *et al.* 2005), the stiffer mechanical response between 50 and 90% internal quadriceps MVF following training does not appear to explain the increased maximal capacity of the MTU for RFD, as octet force at 50 and 100 ms was ~40 and ~62% MVF, respectively. Nevertheless, a stiffer MTU at high mechanical loads is thought to reduce the risk of soft tissue injury (Garrett, 1990; Buchanan & Marsh, 2002), so this adaptation after merely 4 weeks of training for explosive strength may have important implications for injury prevention.

Despite the increase in MTU stiffness at high force levels, there was no change in MTU stiffness at low force levels. This may be due to the poor reliability of MTU stiffness at low force levels (ICC = 0.29), although the consistent MTU stiffness measurements in the untrained leg tend to negate this explanation. It is possible that the separate components of the MTU may differentially contribute to MTU stiffness as force increases and could show distinct mechanical responses to the training (e.g. increased tendon stiffness at high force and unchanged muscle stiffness at low force). Further research is required, using measurements at both the aponeurosis and the tendon, to investigate the possibility of a differential mechanical response of the separate components of the MTU to strength training.

The only measured effect in the untrained leg was a small (4%) increase in MVF. This effect provides evidence of neural adaptations that may have contributed to the improvements in MVF in both the trained and the untrained legs. Given the apparent importance of neural factors for explosive force production (de Ruiter *et al.* 2004; Tillin *et al.* 2010) some cross-over of explosive performance to the untrained leg might also have been expected, but we found no evidence for this with the present training intervention. There was also no change

in any of the twitch force parameters of the trained or untrained legs, which is consistent with the equivocal evidence for changes in twitch force parameters following any type of strength training (Duchateau & Hainaut, 1984; Brown *et al.* 1990; Shima *et al.* 2002; Pucci *et al.* 2006; Del Balso & Cafarelli, 2007; Lee *et al.* 2009). Considering the significant improvements in octet  $F_{50}$  and  $F_{100}$  in the present study, it could be that the mechanical response to a single supramaximal electrical impulse is insensitive to subtle adaptations of the MTU. Alternatively, changes in excitation–contraction coupling and calcium transients after training could differentially influence twitch and octet responses, although this possibility requires further investigation. Finally, there was no change in voluntary or involuntary EMD<sub>max</sub> after training, which supports earlier evidence that EMD is not influenced by the explosive force capacity of the neuromuscular system (Tillin *et al.* 2010).

In conclusion, training for explosive, rather than maximal, force production improved voluntary and involuntary explosive force, maximal strength and MTU stiffness at high force levels after only 4 weeks. Increased agonist neural drive and maximal strength accounted for improved voluntary explosive force production in the early and late phases, respectively. The capacity of the MTU for explosive force production in response to an evoked octet increased in proportion to maximal strength, suggesting a common peripheral mechanism that may be hypertrophy. The stiffer response of the MTU at high force levels following training was unlikely to contribute to the increased octet explosive force, but may have important implications for injury prevention.

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